INCH-POUND
MIL-DTL-24643/38E
22 August 2002
SUPERSEDING
MIL-C-24643/38D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSECM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, in accordance with ASTM B 286, AWG Size No. 20,7 strands, tin coated.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
Fourth -	Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form each a pair.
Fifth -	Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
Sixth -	Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent

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Construction (watertight) (continued)

Seventh - Shield insulation of two polyester tapes per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

Groups of 7 (8 groups)

First - Copper conductor, AWG No. 18, class B stranding, (7 by 0.0152 inch) uncoated. Maximum diameter of 0.05 inch.

Second - Ethylene propylene rubber or cross-linked polyethylene insulation. Nominal wall thickness 0.008 inch. Colored insulation five black and two white conductors for each group.

Third - Seven conductors, five black and two white, with the white conductors adjacent in the layer of six, cabled with a lay not greater than 2 inches, to form each group.

Fourth - Two polyester tapes, per NEMA FI 4, outer tape sealed, an alternate shield insulation may consist of one polyester tape per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

ASSEMBLY:

First - Five pairs cable to form a core. Three pairs and eight groups of seven cabled over the core, with a lay not greater than 24 times the pitch diameter of the layer. Pair nos. 1 through 5 in core; the sequence of the outer layer - pair no. 6, groups nos. 1, 2 and 3; pair no. 7; groups nos. 4, 5 and 6; pair No. 8; group nos. 7 and 8. Fillers shall be used as necessary to form a well rounded assembly.

Second - Binder tape applied helically with overlap.

Construction variant for type LSECM

Third - Cross-linked polyolefin jacket. Cable jacket thickness 0.075 inch minimum average.

NOTE: 1. Variant LSECM completed cable diameters: Minimum 1.290 inches, maximum 1.370 inches. Military part number M24643/38-01UN.

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EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor resistance - ohms/1000 feet at 25 °C, maximum	
ASTM B 286, size 20-7	11.03
AWG no 18 Class B	7.47
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Shield to shield	500
Insulation resistance - Megohms -1000 feet minimum	
Conductor to shield	500
Component shield to shield	100
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-DTL-24643 for limits of water leakage	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum	30
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200 °C	50
Tear - pounds per inch thickness, minimum ASTM D 470	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75

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Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125
Cable sealant removability ..... No failure

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation group covering	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80

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Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS:
1000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-36)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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